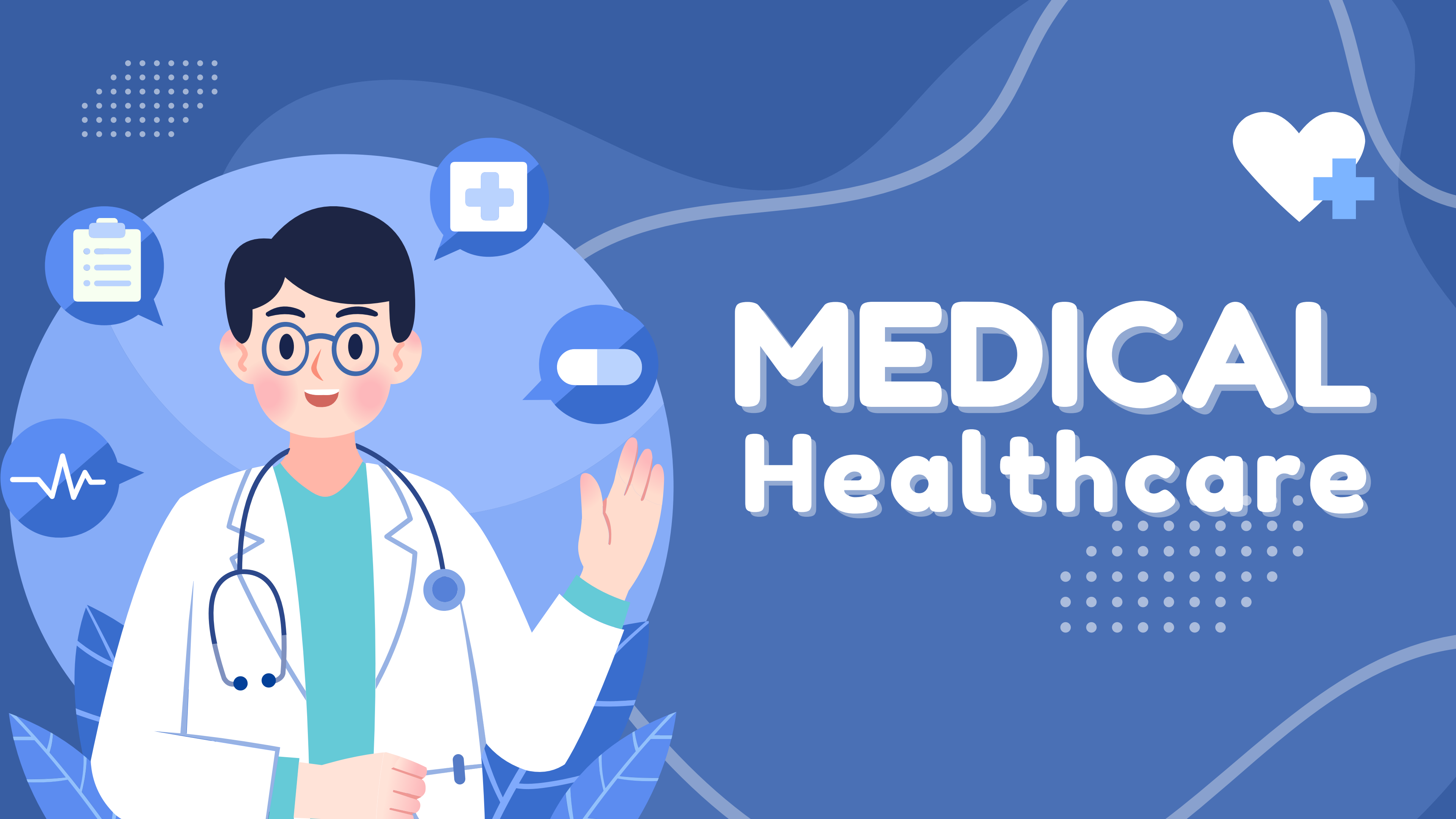




MEDICAL Healthcare

HEALTHCARE For Patient



14K

Sum of length_of_stay

48K

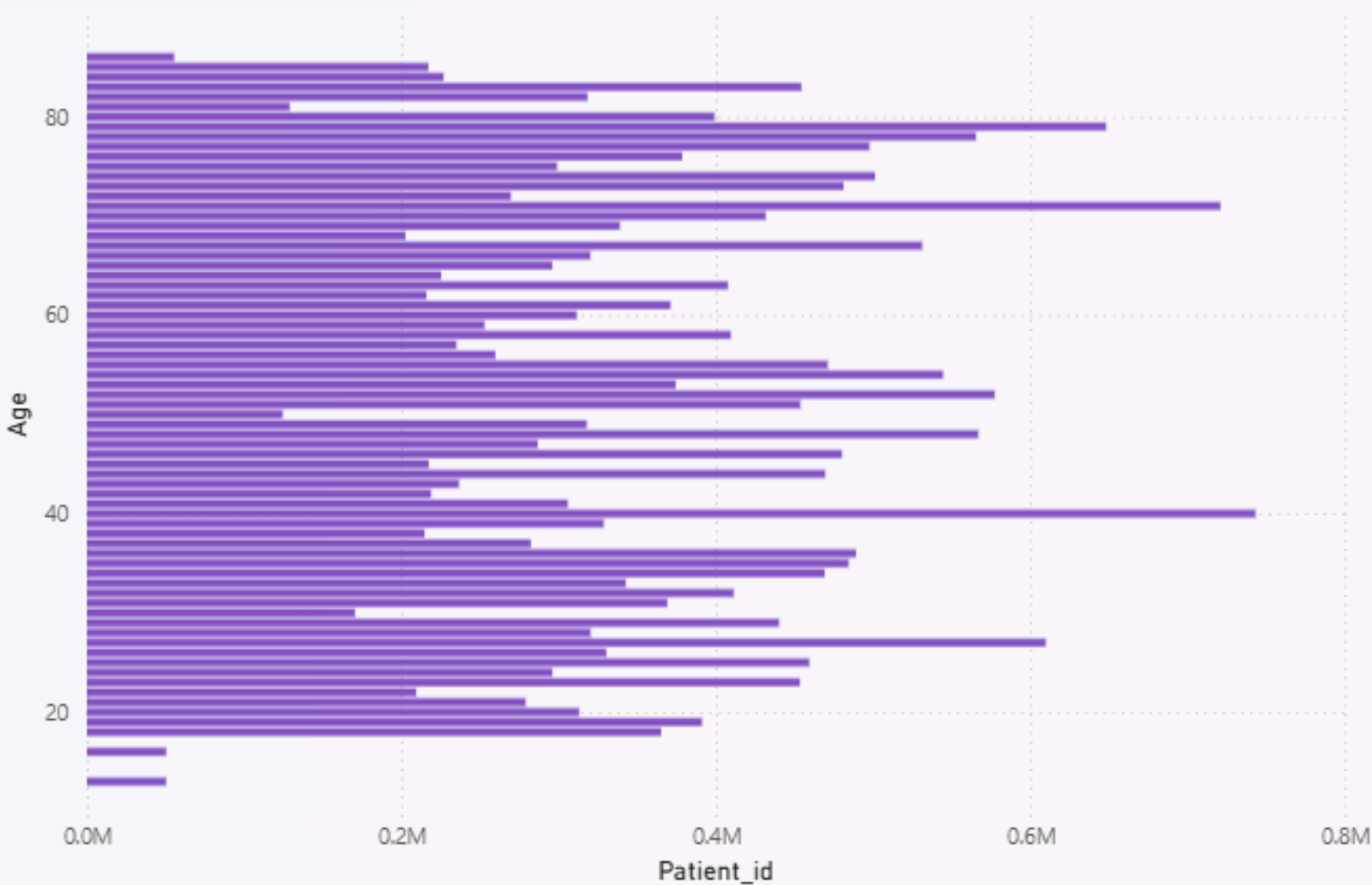
Sum of Age

25M

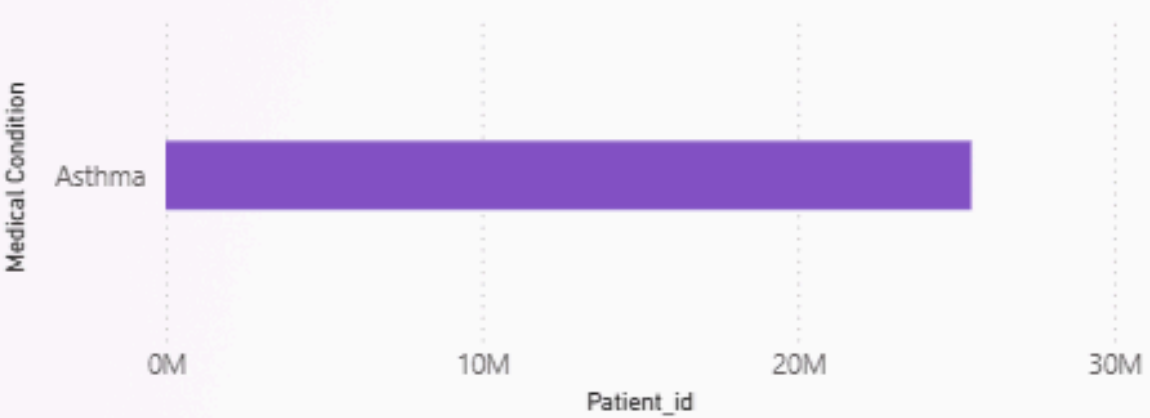
Sum of Patient_id

Gender	Medical Condition	Insurance Provider
> ☐ Female	☐ Arthritis	☐ Aetna
	☒ Asthma	☒ Blue Cross
	☐ Cancer	☐ Cigna
> ☒ Male	☐ Diabetes	☐ Medicare
	☐ Hypertention	☐ UnitedHealthcare
	☐ Obesity	

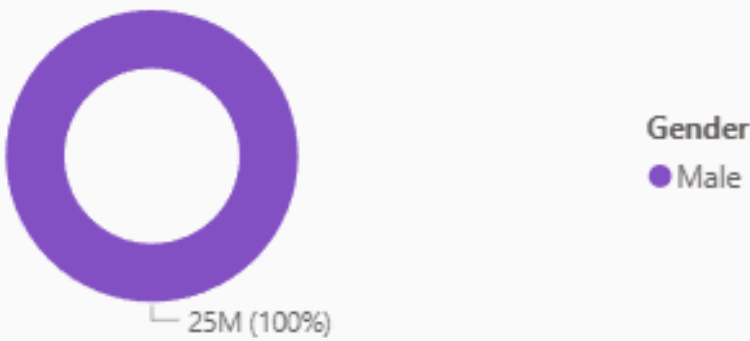
Patient_id by Age



Patient_id by Medical Condition

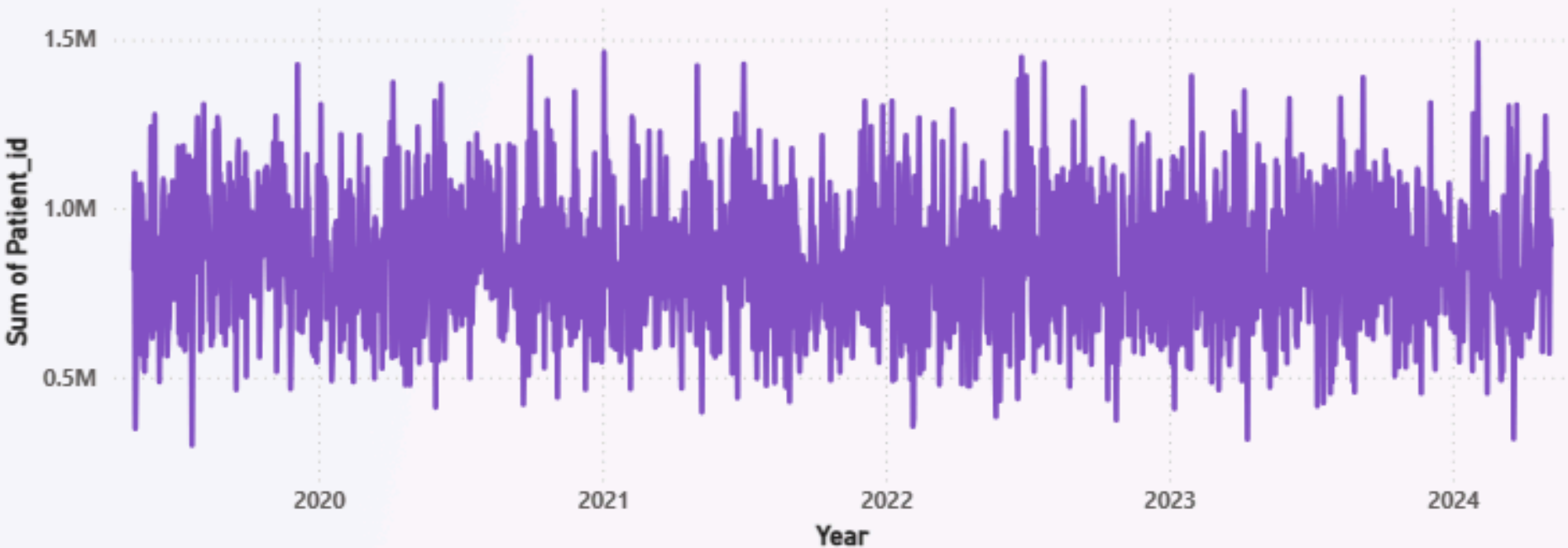


Patient_id by Gender

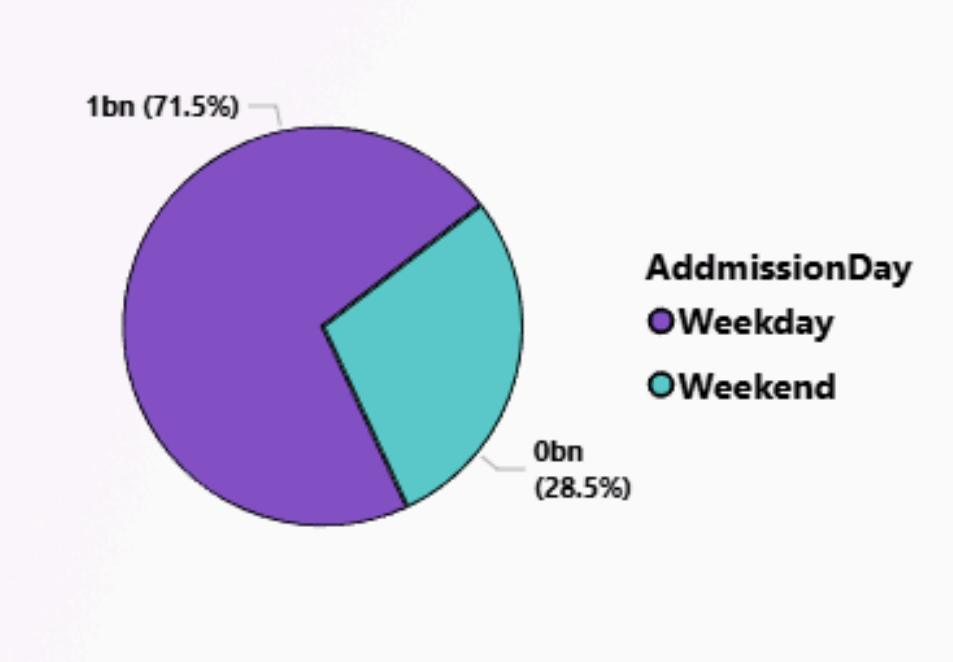


HEALTHCARE For hospital vs Operation

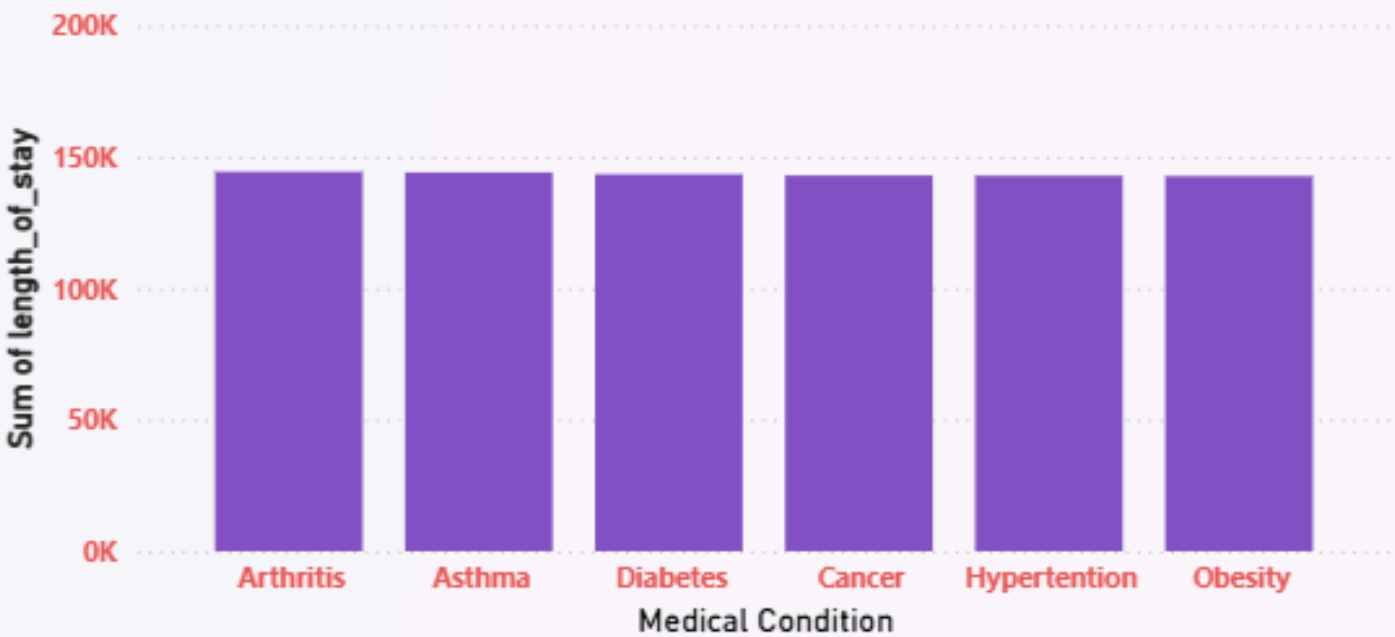
Sum of Patient_id by Year, Quarter, Month and Day



Sum of Patient_id by AdmissionDay



Sum of length_of_stay by Medical Condition



Season	Arthritis	Asthma	Cancer	Diabetes	Hypertention	Obesity	Total
Autumn	63383539	60939350	65663711	65297681	63410793	64358127	383053201
Spring	61153348	62900910	64477482	64248255	64392330	64806758	381979083
Summer	66908057	67519035	65415800	68343952	64757337	65092187	398036368
Winter	65081014	63180640	60718198	61955331	62757982	63390933	377084098
Total	256525958	254539935	256275191	259845219	255318442	257648005	1540152750



32.64K✓
Goal: 32.64K (+0%)

1.08M!

Goal: 23.52M (-95.42%)

A bar chart titled 'Sum of Billing Amount' on the y-axis and 'Insurance Provider' on the x-axis. The y-axis has major ticks at 0.0bn, 0.1bn, 0.2bn, and 0.3bn. The x-axis lists five insurance providers: Cigna, Medicare, Blue Cross, UnitedHealthc..., and Aetna. All five bars are purple and have a height of approximately 0.28bn.

Insurance Provider	Sum of Billing Amount (bn)
Cigna	0.28
Medicare	0.28
Blue Cross	0.28
UnitedHealthc...	0.28
Aetna	0.28

Billing_level

- high
- medium
- low

Billing_level	Count	Percentage
high	1bn	71.37%
medium	0bn	20.35%
low	0bn	8.28%

Patient_id	Age	Billing Amount	Medical Condition	Year	Quarter	Month	Day	Insurance Provider	Sum of length_of_stay
37839	51	52,764.28	Hypertention	2023	Qtr 3	September	8	Blue Cross	26
34582	79	52,373.03	Cancer	2021	Qtr 2	June	19	UnitedHealthcare	14
51390	74	52,373.03	Cancer	2021	Qtr 2	June	19	UnitedHealthcare	14
5737	22	52,271.66	Hypertention	2019	Qtr 4	November	2	Blue Cross	9
7190	18	52,211.85	Diabetes	2020	Qtr 3	August	11	Aetna	1
1731	42	52,181.84	Asthma	2021	Qtr 4	November	10	Aetna	7
6704	42	52,170.04	Arthritis	2021	Qtr 2	June	27	Cigna	2
52741	43	52,170.04	Arthritis	2021	Qtr 2	June	27	Cigna	2
33050	53	52,154.24	Cancer	2019	Qtr 2	May	30	UnitedHealthcare	23
34580	38	52,102.24	Cancer	2019	Qtr 2	May	20	Cigna	9
13794	80	52,092.67	Cancer	2023	Qtr 4	October	15	Medicare	2
38201	21	52,024.73	Obesity	2022	Qtr 3	September	2	Cigna	26
10936	54	51,986.08	Diabetes	2023	Qtr 3	August	14	Medicare	21
32767	20	51,975.97	Arthritis	2023	Qtr 2	April	22	Blue Cross	27
Total		860750							

Limit to 1000 rows															
1 • create database health; 2 • rename table healthcare_dataset to hlth; 3 • 4 • select * from hlth;															
Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:															
	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
▶	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281305978155	328	Urgent	2024-02-02	Paracetamol	Normal
	LesLie TERy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327286577885	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
	DaNnY sMith	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096078842456	205	Emergency	2022-10-07	Aspirin	Normal
	andrEw waTtS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.78240987528	450	Elective	2020-12-18	Ibuprofen	Abnormal
	adrIENNE bELL	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317813937623	458	Urgent	2022-10-09	Penicillin	Abnormal
	EMILY JOHNSOn	36	Male	A+	Asthma	2023-12-20	Taylor Newton	Nunez-Humphrey	UnitedHealthcare	48145.11095104189	389	Urgent	2023-12-24	Ibuprofen	Normal
	edwArD EDWaRDs	21	Female	AB-	Diabetes	2020-11-03	Kelly Olson	Group Middleton	Medicare	19580.87234486093	389	Emergency	2020-11-15	Paracetamol	Inconclusive
	CHrisTInA MARTinez	20	Female	A+	Cancer	2021-12-28	Suzanne Thomas	Powell Robinson and Valdez,	Cigna	45820.46272159459	277	Emergency	2022-01-07	Paracetamol	Inconclusive
	JASmINe aGuIlAR	82	Male	AB+	Asthma	2020-07-01	Daniel Ferguson	Sons Rich and	Cigna	50119.222791548505	316	Elective	2020-07-14	Aspirin	Abnormal
	ChRISTopher BerG	58	Female	AB-	Cancer	2021-05-23	Heather Day	Padilla-Walker	UnitedHealthcare	19784.63106221073	249	Elective	2021-06-22	Paracetamol	Inconclusive
	mIchElLe daniELs	72	Male	O+	Cancer	2020-04-19	John Duncan	Schaefer-Porter	Medicare	12576.795609050234	394	Urgent	2020-04-22	Paracetamol	Normal
	aaRon MARTiNeZ	38	Female	A-	Hypertension	2023-08-13	Douglas Mayo	Lyons-Blair	Medicare	7999.586879604188	288	Urgent	2023-09-05	Lipitor	Inconclusive
	connOR HANsEn	75	Female	A+	Diabetes	2019-12-12	Kenneth Fletcher	Powers Miller, and Flores	Cigna	43282.28335770435	134	Emergency	2019-12-28	Penicillin	Abnormal
	rObErT bAuer	68	Female	AB+	Asthma	2020-05-22	Theresa Freeman	Rivera-Gutierrez	UnitedHealthcare	33207.706633729606	309	Urgent	2020-06-19	Lipitor	Normal
	bROOkE brady	44	Female	AB+	Cancer	2021-10-08	Roberta Stewart	Morris-Arellano	UnitedHealthcare	40701.599227308754	182	Urgent	2021-10-13	Paracetamol	Normal
	MS. nAtAlIE gAMble	46	Female	AB-	Obesity	2023-01-01	Maria Dougherty	Cline-Williams	Blue Cross	12263.357425021362	465	Elective	2023-01-11	Aspirin	Inconclusive
	haley perkins	63	Female	A+	Arthritis	2020-06-23	Erica Spencer	Cervantes-Wells	UnitedHealthcare	24499.847903736576	114	Elective	2020-07-14	Paracetamol	Normal
	mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	2020-03-08	Justin Kim	Torres, and Harrison Jones	Cigna	17440.465444124675	449	Urgent	2020-04-02	Paracetamol	Abnormal
	LADY B. B. B.	34	Female	A-	Hypertension	2021-02-04	Justin Moore Jr	Houston PLC	Blue Cross	10043.02301302416	260	Elective	2021-02-14	Aspirin	Abnormal

```

63 -- feature engineering for the column (date of admission , discharge date i.e to convert to date since it's a string )
64 -- column , age , billing amount and
65
66 • select `date of admission`, `discharge date` ,
67    datediff(
68        str_to_date(`discharge date`, '%Y-%m-%d'),
69        str_to_date(`date of admission`, '%Y-%m-%d')
70    )as length_of_stay
71 from hlth ;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows: 

	date of admission	discharge date	length_of_stay
▶	2024-01-31	2024-02-02	2
	2019-08-20	2019-08-26	6
	2022-09-22	2022-10-07	15
	2020-11-18	2020-12-18	30
	2022-09-19	2022-10-09	20
	2023-12-20	2023-12-24	4
	2020-11-03	2020-11-15	12
	2021-12-28	2022-01-07	10
	2020-07-01	2020-07-14	13
	2021-05-23	2021-06-22	30
	2020-04-19	2020-04-22	3
	2023-08-13	2023-09-05	23
	2019-12-12	2019-12-28	16
	2020-05-22	2020-06-19	28
	2021-10-08	2021-10-13	5
	2023-01-01	2023-01-11	10

```
73 • select age ,
74     case
75         when age < 18 then 'Child'
76         when age between 18 and 30 then 'Young_Adult'
77         when age between 31 and 50 then 'Adult'
78         when age between 51 and 70 then 'senior'
79         else 'elderly'
80     end as AgeGroup
81 from hlth ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	age	AgeGroup
▶	30	Young_Adult
	62	senior
	76	elderly
	28	Young_Adult
	43	Adult
	36	Adult
	21	Young_Adult
	20	Young_Adult
	82	elderly
	58	senior
	72	elderly
	38	Adult
	75	elderly
	68	senior
	44	Adult
	46	Adult

```

82
83 • select `billing Amount`,
84 case
85     when `billing amount` < 5000 then 'low'
86     when `billing amount` between 5000 and 15000 then 'medium'
87     else 'high'
88 end as Billing_level
89 from hlth ;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows: 

	billing Amount	Billing_level
▶	18856.281305978155	high
	33643.327286577885	high
	27955.096078842456	high
	37909.78240987528	high
	14238.317813937623	medium
	48145.11095104189	high
	19580.87234486093	high
	45820.46272159459	high
	50119.222791548505	high
	19784.63106221073	high
	12576.795609050234	medium
	7999.586879604188	medium
	43282.28335770435	high
	33207.706633729606	high
	40701.599227308754	high
	12263.357425021362	medium

Limit to 1000 rows

```
91 -- looking for the admission of patient in which season , weekend
92
93 • select `date of admission`,
94 case
95     when month(`date of admission`) in (1,2,12) then 'Winter'
96     when month(`date of admission`) in (3,4,5) then 'Spring'
97     when month(`date of admission`) in (6,7,8) then 'Summer'
98     else 'Autumn'
99 end as season from hlth ;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Fetch rows:



	date of admission	season
▶	2024-01-31	Winter
	2019-08-20	Summer
	2022-09-22	Autumn
	2020-11-18	Autumn
	2022-09-19	Autumn
	2023-12-20	Winter
	2020-11-03	Autumn
	2021-12-28	Winter
	2020-07-01	Summer
	2021-05-23	Spring
	2020-04-19	Spring
	2023-08-13	Summer
	2019-12-12	Winter

```
100
101 • select `date of admission`,
102     case
103         when dayofweek(`date of admission`) in (1,7) then 'Weekend'
104         else 'Weekday'
105     end as AdmissionDay from hlth;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	date of admission	AdmissionDay
▶	2024-01-31	Weekday
	2019-08-20	Weekday
	2022-09-22	Weekday
	2020-11-18	Weekday
	2022-09-19	Weekday
	2023-12-20	Weekday
	2020-11-03	Weekday
	2021-12-28	Weekday
	2020-07-01	Weekday
	2021-05-23	Weekend
	2020-04-19	Weekend

```

107  -- since the hlth table don't have a ID
108  •  alter table hlth
109      add column Patient_id int not null auto_increment primary key first ;
110
111  •  select * from hlth ;

```

Result Grid |   Filter Rows: | Edit:    | Export/Import:   | Wrap Cell Content:  | Fetch rows:

	Patient_id	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insu Prov
▶	1	65	Male	O+	Cancer	2022-10-12	Daniel Murphy	Group Armstrong	Medic
	2	54	Female	AB+	Cancer	2020-03-04	Misty Garcia	Russell Ltd	Unite
	3	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue C
	4	58	Female	A-	Obesity	2021-02-14	Doris Hawkins	and Ramirez Figueroa May,	Medic
	5	57	Male	B-	Arthritis	2021-08-01	Melissa Fernandez	Jones Ltd	Cigna
	6	38	Female	AB+	Asthma	2020-01-24	Kari Moore	Gordon-Ward	Medic
	7	54	Male	O+	Obesity	2021-07-30	David Hubbard	Barnes and Sons	Medic
	8	26	Female	A-	Hypertention	2019-11-10	Caleb James	Michael-Jenkins	Medic
	9	31	Female	O+	Hypertention	2021-12-01	Emily Patterson	Ltd Schwartz	Cigna
	10	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medic
	11	42	Male	A+	Diabetes	2024-01-04	Lonnie Morgan	Levine-Miller	Medic
	12	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna
	13	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Va...	Medic
	14	58	Male	A-	Diabetes	2020-05-04	Heather Smith	Nelson-Phillips	Cigna
	15	22	Male	O+	Hypertention	2023-09-09	John Smith	Knight and Nguyen, Riggs	Unite
	16	22	Male	O+	Obesity	2023-04-16	Patricia Moore	Hardy Inc	Unite
	17	29	Female	B+	Asthma	2020-04-25	Matthew Gray	Turner, and Williams Merc...	Cigna
	18	42	Female	AB+	Cancer	2022-09-10	Kathleen Hagan	White, White	Aetna

THANK
YOU ♥ +

